SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Shaik Asha(192473025) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i.Map how each affects the Data structure and Security level
        ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 24	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

99
100

CLASSICAL ENCRYPTION SCHEMES - SUBSTITUTION vs TRANSPOSITION

CLASSICAL ENCRYPTION SCHEMES

Substitution Techniques
(Replace characters with other characters)

Examples
- Caesar Cipher
- Monoalphabetic
- Playfair Cipher
- Vigenère Cipher

Effect on Data Structure
Characters change
Order generally preserved

Security Level
Frequency of letters preserved
Vulnerable to frequency analysis

Transposition Techniques
(Rearrange characters without changing)

Examples
- Rail Fence Cipher
- Route Cipher
- Columnar Transposition

Effect on Data Structure
Characters preserved
Order changed

Security Level
Frequency of letters preserved
Vulnerable to permutation/ pattern analysis

Which is more vulnerable & Why?

Substitution (especially monoalphabetic)
→ Preserves letter frequency
→ Easily broken using frequency analysis
Example: Repeated 'X' may indicate a common letter

Transposition
→ Does not change character frequency
→ Vulnerable to anagram/pattern analysis
Example: Same letters, different order

Key Difference and Similarities

Feature	Substitution	Transposition
What Changes?	characters	Positions
Characters Preserved?	No	Yes
Order Preserved?	Generally yes	No
Frequency Preserved	Yes	Yes
Main attack	Frequency analysis	Permutation/ Pattern analysis
Modern Security	Weak	Weak

Similarities:
→ Both convert plaintext to ciphertext
→ Both use a secret key/rule
→ Both aim to hide the original message
→ Each are reversible
→ Both are vulnerable to cryptanalysis

Conclusion:
→ Both are weak by modern standards, But simple substitution is particularly vulnerable to frequency analysis, while transposition is vulnerable to permutation.

SECURITY ATTACKS, THREATS, RISKS AND SECURITY GOALS

Security Attacks → exploit → Threats → create → Risks → affect → Security Goals (CIA Triad)

Phishing	Credential theft	Unauthorized account access	Confidentiality	Example: Fake email steals login credentials (banking)
DDoS	Service flooding	Service unavailable	Availability	Example: Mirai botnet causes website outage
SQL injection	Database access/ modification	Data theft / manipulation	Confidentiality + Integrity	Example: Malicious SQL command exposes customer data
Man-in-the Middle (MITM)	Interception or modification of communication	Data disclosure (or) tampering	Confidentiality + integrity	Example: Attacker intercepts data on public Wi-Fi
Ransomware	File Encryption/ data destruction	Loss of data access	Availability	Example: WannaCry encrypts files and disrupts systems

SECURITY GOALS (CIA Triad)

Confidentiality
- Prevents unauthorized disclosure
- **Example Control:** Encryption access control

Integrity
- Prevents unauthorized modification
- Example Control: Hashing digital signatures

Availability
- Ensures access to data/ systems
- Example Control: Backups, redundancy, DDoS protection

Overall Relationship

Attack → Exploits Vulnerability → Creates threat → Produces Risk → Causes Security goal validation